LESSON PLAN

SFNA (MOD AIRMANSHIP) (24)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various types of hydraulic fluids, anti-corrosion compounds, deicing and anti-freezing compounds, coolant liquids, water injection mixtures
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various types of hydraulic fluids, anti-corrosion compounds, deicing and anti-freezing compounds, coolant liquids, water injection mixtures, cooling oils, transformer oils, flushing oil, corrosion inhibitors etc.
5. Specific Objectives	24.1.3.1 Trainee will be able to identify various types of greases, fluids and compounds.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of fluid samples using test strips and visual inspection; Safety: Wear gloves and eye protection; Supervision: Instructor monitors sample handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of hydraulic fluid in an aircraft system?
(b) Name one property that hydraulic fluid must possess to prevent corrosion.
(c) Why is it critical to use the correct type of coolant liquid in aircraft engines?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of Hydraulic Fluids • Anti-Corrosion Compounds • Deicing and Anti-Freezing Compounds • Coolant Liquids and Water Injection Mixtures	L & Ds	PPT, IB	15 min
(b)	• Cooling Oils and Transformer Oils • Flushing Oil and Corrosion Inhibitors • Practical Identification of Fluid Samples • Safety Precautions for Fluid Handling	L & Ds	PPT, PDF	20 min
(c)	• Comparison of Fluid Properties • Storage and Handling Requirements • Common Contamination Sources • Impact of Wrong Fluid Selection	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Differentiate between the primary use of hydraulic fluid and transformer oil.
(b) Explain why water injection mixtures require specific anti-freezing properties.
(c) List two safety precautions required when handling anti-corrosion compounds.

12. Home Assignment and Follow Up	Prepare a brief note on the properties and usage of various types of fuels for the next lesson plan (LP 06).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Various types of fuels, their properties and usage. usages of additives	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min